

Multi-Modal Root Cause Analysis in Microservices

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Problem Statement

- Microservice architectures generate massive volumes of metrics, logs, and traces across hundreds of services
- When a failure occurs, it cascades through dependencies, a memory leak in one service causes timeouts, errors, and failures in 5+ downstream services
- Engineers must manually correlate signals across tools to find the root cause
- 82% of organizations have Mean Time To Resolve (MTTR) > 1 hour (up from 47% in 2021)
- 48% cite knowledge gaps as the primary obstacle to faster diagnosis

Goal: Automatically identify the root cause, explain why, and provide calibrated confidence in seconds, not hours



Problem Background

Current Challenges in RCA Solutions

Key Challenges:

- 86% say data volume is beyond human ability to manage ²
- Organizations use average 10 monitoring tools ²
- 51% cite observability as top cloud-native challenge ³

Why Single-Modality Fails:

- Metrics-only: Sees CPU spike, misses memory leak cause
- Logs-only: Sees ERROR, misses thread pool exhaustion
- Traces-only: Sees latency, misses config change

Year	MTTR > 1 HOUR	Change
2021	47%	Baseline
2022	64%	+17%
2023	74%	+10%
2024	82%	+8%

Literature Survey



S. NO	AUTHOR(S), YEAR	METHOD / APPROACH	KEY CONTRIBUTION	GAP IDENTIFIED
1	Robust Failure Diagnosis of Microservice System Through Multimodal Data (DiagFusion) S. Zhang et al., 2023, IEEE TSC	Multi-modal GNN over metrics, logs, and traces; FastText embeddings; data augmentation	First unified multi-modal GNN for microservice fault diagnosis with joint RCA and classification	No causal reasoning; limited interpretability; uncalibrated confidence; requires labeled data
2	FOCAL: Contrastive Learning for Multimodal Time Series Sensing Signals in Factorized Orthogonal Latent Space – S. Liu et al., 2023, NeurIPS	Factorized contrastive learning separating shared/private features	Effective disentanglement of cross-modal vs modality-specific features	Not designed for observability data; no causal discovery; needs paired samples
3	TS-TCC: Self-Supervised Contrastive Representation Learning for Semi-Supervised Time-Series Classification – E. Eldele et al., 2023, IEEE TPAMI	Temporal & contextual contrastive learning with time-series augmentations	Strong temporal representations with only 1% labeled data	Single-modal only; no multimodal fusion or RCA
4	DeepTraLog: Trace-Log Combined Microservice Anomaly Detection through Graph-based Deep Learning – C. Zhang et al., 2022, ICSE	GNN on trace-log graphs; reconstruction-based anomaly detection	First to jointly model traces and logs via graphs	No metrics; anomaly detection only; no RCA
5	Improving Log-Based Anomaly Detection by Pre-Training Hierarchical Transformers (HiLBERT) – W. Huang et al., 2023, IEEE TC	Hierarchical transformer with masked & next-template prediction	Captures multi-level log semantics; good cross-system generalization	Log-only; requires parsing; expensive pre-training

Literature Survey



S. NO	AUTHOR(S), YEAR	METHOD / APPROACH	KEY CONTRIBUTION	GAP IDENTIFIED
6	Root Cause Analysis of Failures in Microservices through Causal Discovery – A. Ikram et al., 2022, NeurIPS	Intervention-aware PC algorithm with hierarchical decomposition	Scalable causal RCA focused on root cause region	Metrics-only; assumes stationarity; weak for nonlinear causality
7	Neural Granger Causality – A. Tank et al., 2022, IEEE TPAMI	Neural networks with group-lasso sparsity for nonlinear GC	Nonlinear causal discovery with theoretical guarantees	Poor scalability; high compute cost
8	Nezha: Interpretable Fine-Grained Root Cause Analysis for Microservices on Multimodal Observability Data – G. Yu et al., 2023, FSE	Event graph mining across multimodal data	Fine-grained, interpretable RCA; strong Top-1 accuracy	Requires fault-free baseline; abstraction may lose detail
9	CHASE: A Causal Hypergraph-based Framework for Root Cause Analysis in Multimodal Microservice Systems – Z. Zhao; Z. Wang; T. Zhang; Z. Shen; H. Dong, 2024	PC-based causal graph + adaptive random walk with feedback	Self-adaptive RCA validated in production	Depends on operator feedback; weak temporal modeling
10	Fault Analysis and Debugging of Microservice Systems: Industrial Survey, Benchmark System, and Empirical Study – X. Zhou et al., 2021, IEEE TSE	Multimodal encoders + attentive heterogeneous graph; causal hypergraph with hypergraph convolution	First causal hypergraph-based multimodal RCA; models higher-order causal relations; outperforms DiagFusion	Supervised training; static causal structure; limited model depth

Literature Survey



S. NO	AUTHOR(S), YEAR	METHOD / APPROACH	KEY CONTRIBUTION	GAP IDENTIFIED
11	Revisiting Confidence Estimation: Towards Reliable Failure Prediction – F. Li et al., 2023, IEEE TPAMI	Auxiliary confidence model with calibration targets	Addresses overconfidence in predictions	Classification-focused; not RCA-oriented
12	A Survey on Explainable Anomaly Detection – Z. Li et al., 2023, ACM TKDD	Systematic literature review	First comprehensive explainability taxonomy	Survey only; no microservice focus
13	Deep Learning for Time Series Anomaly Detection: A Survey – Z. Choi et al., 2023, ACM CSUR	Survey of 100+ DL-based AD methods	Clear taxonomy; highlights interpretability gaps	Survey only; limited practical guidance
14	CloudRanger: Root Cause Identification for Cloud Native Systems – P. Wang et al., 2018, IEEE CCGRID	PC-based causal graph + second-order random walk	Early cloud-native RCA framework	Metrics-only; sensitive to graph quality
15	Confidence Estimation via Auxiliary Models – C. Corbière et al., 2022, IEEE TPAMI	Auxiliary network for true-class probability	Principled confidence estimation	Extra training cost; not AIOps-specific

Literature Survey



S. NO	AUTHOR(S), YEAR	METHOD / APPROACH	KEY CONTRIBUTION	GAP IDENTIFIED
16	GAIA: Generic AIOps Atlas Dataset – CloudWise Team, 2021, GitHub	Simulated system with metrics, logs, traces	First open multimodal AIOps dataset	Simulated; limited fault diversity
17	A Comprehensive Survey on Root Cause Analysis in (Micro)Services: Methodologies, Challenges, and Trends – Y. Zhang et al., 2024, arXiv	RCA survey by modality and technique	Identifies open research gaps	Survey only; rapidly evolving field
18	Enjoy Your Observability: An Industrial Survey of Microservice Tracing and Analysis – H. Zhao et al., 2022, EMSE	Industry survey on tracing practices	Real-world insights on tracing challenges	Limited generalizability

Existing Solutions and Gaps

- **DiagFusion (Base Paper):**

- Uses Graph Neural Networks on multimodal events (metrics, logs, traces) for fault localization.
- Limitations: Lacks causal reasoning, Struggles with >200 metrics, no explanations

- **MULAN:**

- Applies contrastive learning with random walks over event graphs.
- Limitations: Does not handle traces, supports only pairwise graph relations, and provides no explanations.

- **RUN:**

- Combines Neural Granger Causality with PageRank-based ranking.
- Limitations: Pairwise only, misses cascade patterns

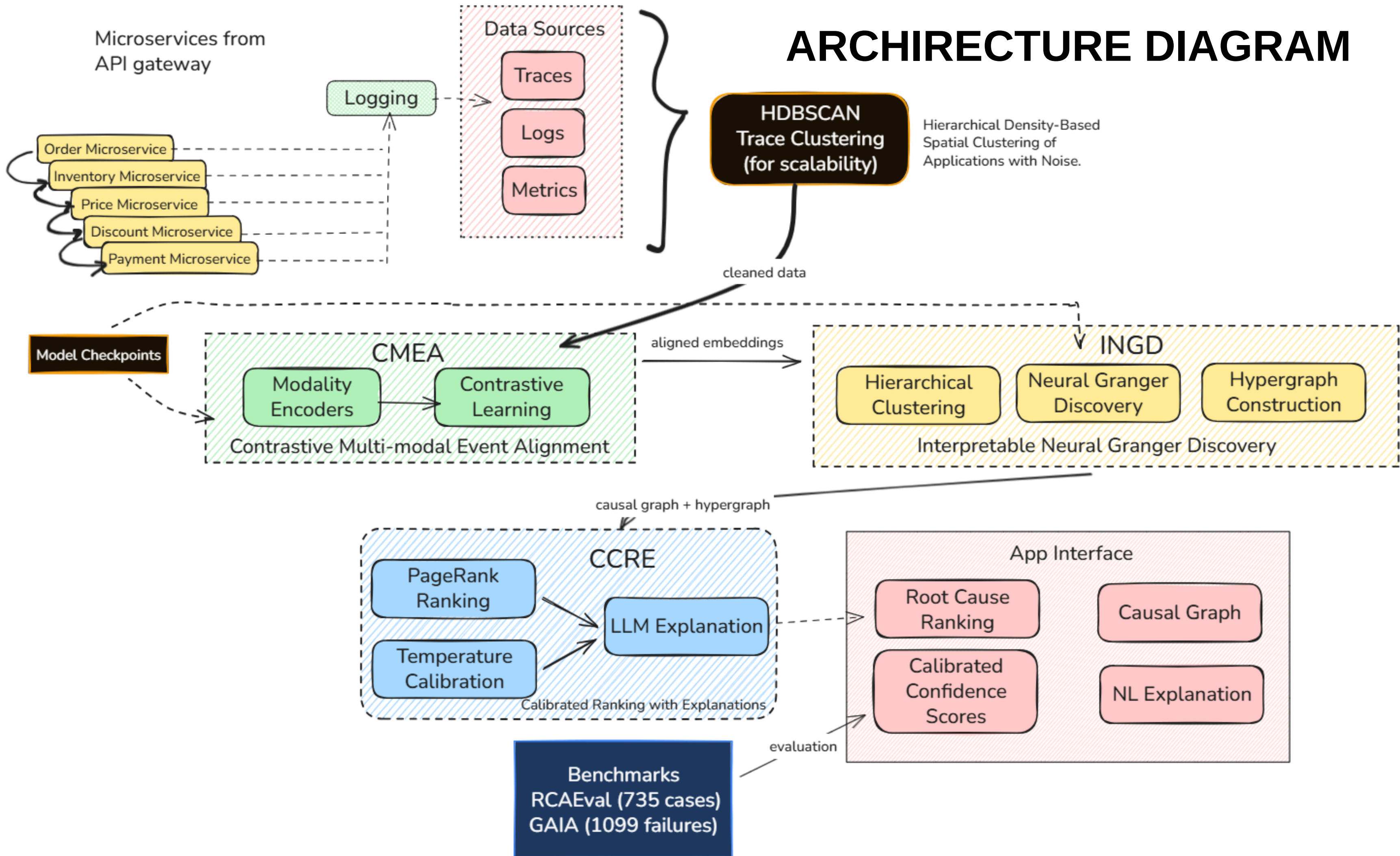
Proposed Solution

Component	Function
CMEA	Contrastive Multi-modal Event Alignment
INGD	Hierarchical Neural Granger Discovery + Hypergraph
CCRE	Calibrated Ranking with Explanations

Key Innovations:

- 1. Scalable causal discovery: Hierarchical divide-and-conquer approach
- 2. Hypergraph structure: Captures multi-way cascading failures
- 3. Human-Friendly Explanations: Domain-specific, historically grounded

ARCHIRECTURE DIAGRAM



Proposed Solutions

CMEA — Contrastive Multi-modal Alignment

- Converts logs to time-series and contrastive learning, extracts modality-invariant features (shared across metrics/logs/traces)
- Why Contrastive Learning?: - Learns what's common across modalities (shared fault signatures) and preserves what's unique (modality-specific evidence)

INGD — Scalable Causal Discovery

- Problem: Neural Granger fails on large systems (200+ metrics)
- Solution: Hierarchical Divide and Conquer (source from RCD) to improve scalability
- Hypergraph Innovation: Regular edge: $A \rightarrow B$ (pairwise only); Hyperedge: $\{A, B, C, D\}$ (captures entire cascade)

CCRE — Calibrated Explanations

- PageRank Ranking : Personalized random walk on hypergraph
- Temperature Scaling: Calibrate confidence (94% means ~94% accurate)
- Backtracks through the causal graphs to identify failure points

Dataset Review

- **RCAEval**

<https://github.com/phamquiluan/RCAEval>

Dataset	System	Cases	Fault Types	Metrics	Logs	Traces
RE1-OB	Online Boutique	125	cpu, mem, disk, delay, loss	49-59	N/A	N/A
RE1-SS	Sock Shop	125	cpu, mem, disk, delay, loss	57-63	N/A	N/A
RE1-TT	Train Ticket	125	cpu, mem, disk, delay, loss	198-238	N/A	N/A
RE2-OB	Online Boutique	90	cpu, mem, disk, delay, loss, socket	69-77	Yes	Yes
RE2-SS	Sock Shop	90	cpu, mem, disk, delay, loss, socket	74-82	Yes	Yes
RE2-TT	Train Ticket	90	cpu, mem, disk, delay, loss, socket	340-376	Yes	Yes
RE3-OB	Online Boutique	30	f1, f2, f3, f4, f5	68-101	Yes	Yes
RE3-SS	Sock Shop	30	f1, f2, f3, f4, f5	80-107	Yes	Yes
RE3-TT	Train Ticket	30	f1, f2, f3, f4, f5	294-322	Yes	Yes

Benchmark includes nine datasets organized into three benchmark suites (RE1, RE2, RE3), each covering three microservice systems (Online Boutique, Sock Shop, Train Ticket). Together, these datasets feature 735 failure cases with 11 fault types

Project Scope

In Scope:

- Multi-modal data ingestion (metrics, traces, logs)
- HDBSCAN trace clustering for scalability
- Contrastive embedding alignment (CMEA)
- Hierarchical Neural Granger causal discovery (INGD)
- Hypergraph construction for multi-way causality
- PageRank ranking with confidence calibration (CCRE)
- LLM explanations
- Evaluation on RCAEval (735 cases) and GAIA dataset

Out of Scope:

- Real-time streaming analysis (we use batch/offline analysis)
- Auto-remediation (we diagnose, not fix)
- Commercial tool API integrations

Timeline and Milestone

Month 1 (Weeks 1–4): Foundation & Setup

- Literature review on RCA, trace clustering, and causal discovery.
- Finalization of datasets and benchmark selection.
- Environment setup and baseline implementation planning.

Outcome:

- ✓ Literature review completed
- ✓ Datasets and benchmarks finalized

Month 2 (Weeks 5–8): Representation & Causal Modeling

- HDBSCAN-based trace clustering.
- CMEA-based scalable embedding module implementation
- INGD: Hierarchical Neural Granger + Hypergraph causal discovery.

Outcome:

- ✓ Scalable embedding module
- ✓ Causal discovery module

Month 3 (Weeks 9–12): Ranking, Explanation & Evaluation

- CCRE: PageRank-based ranking with calibration
- Root cause explanation generation
- Integration of all modules
- Comparison with 15+ state-of-the-art baselines

Outcome:

- ✓ Ranking & explanation module
- ✓ Comprehensive performance metrics

Month 4 (Weeks 13–16): Application & Finalization

- End-to-end application integration
- System optimization and ablation studies
- Documentation and report writing
- Final presentation preparation

Outcome:

- ✓ Working application
- ✓ Final report and presentation

Tools and Technologies

Category	Technologies
Languages	Python 3.10+, TypeScript, Rust
ML Frameworks	PyTorch, scikit-learn
Embeddings	FastText, Sentence-BERT, Drain (log parsing)
Causal Discovery	gCastle, React Flow, Neural Granger
Benchmarks	RCAEval (735 cases), GAIA, Train- Ticket
Observability	OpenTelemetry, Jaeger

References

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Thank
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